

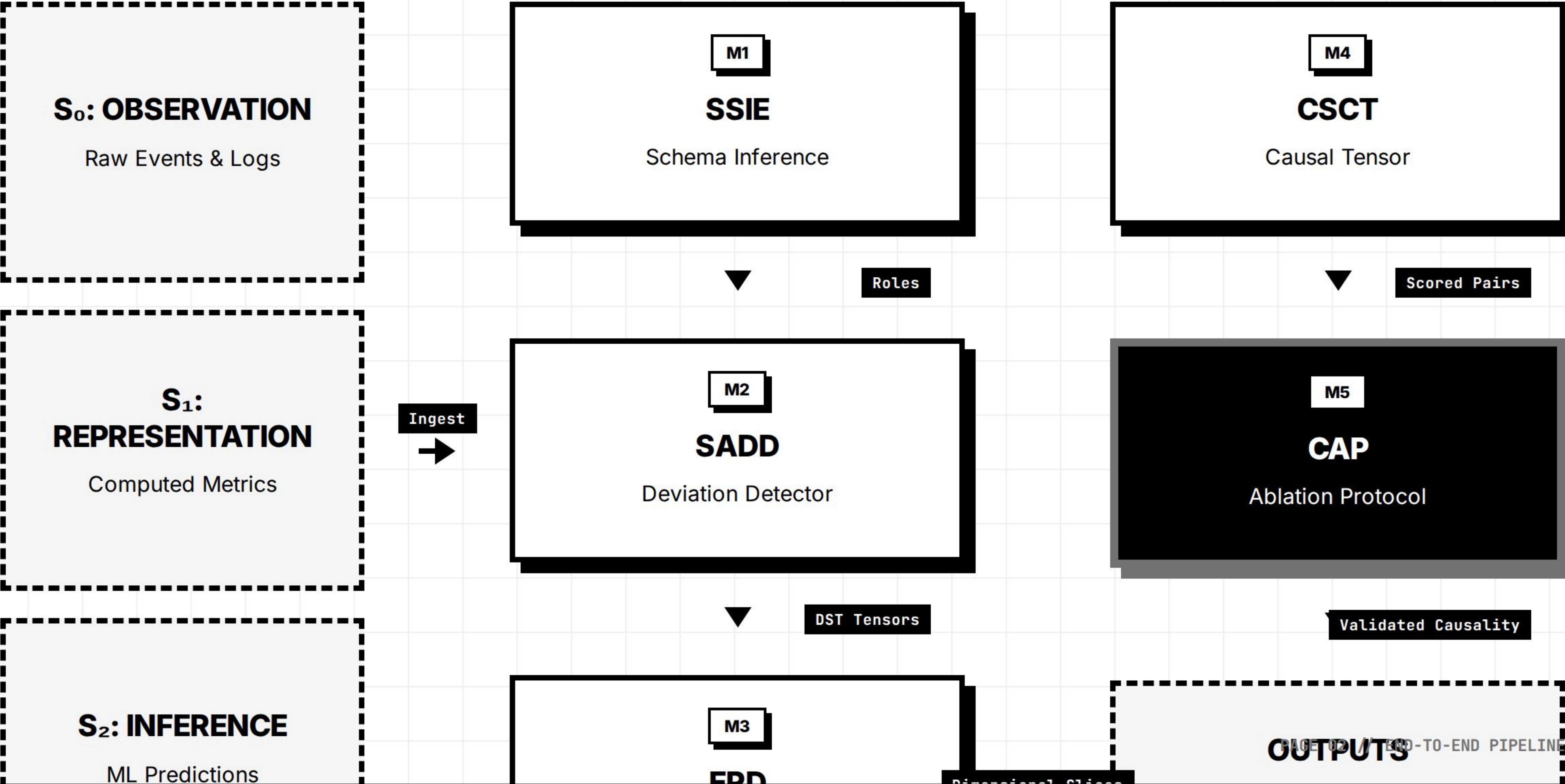
STRICTLY CONFIDENTIAL // PATENT ARCHITECTURE

AMACE COMPONENT BLUEPRINT

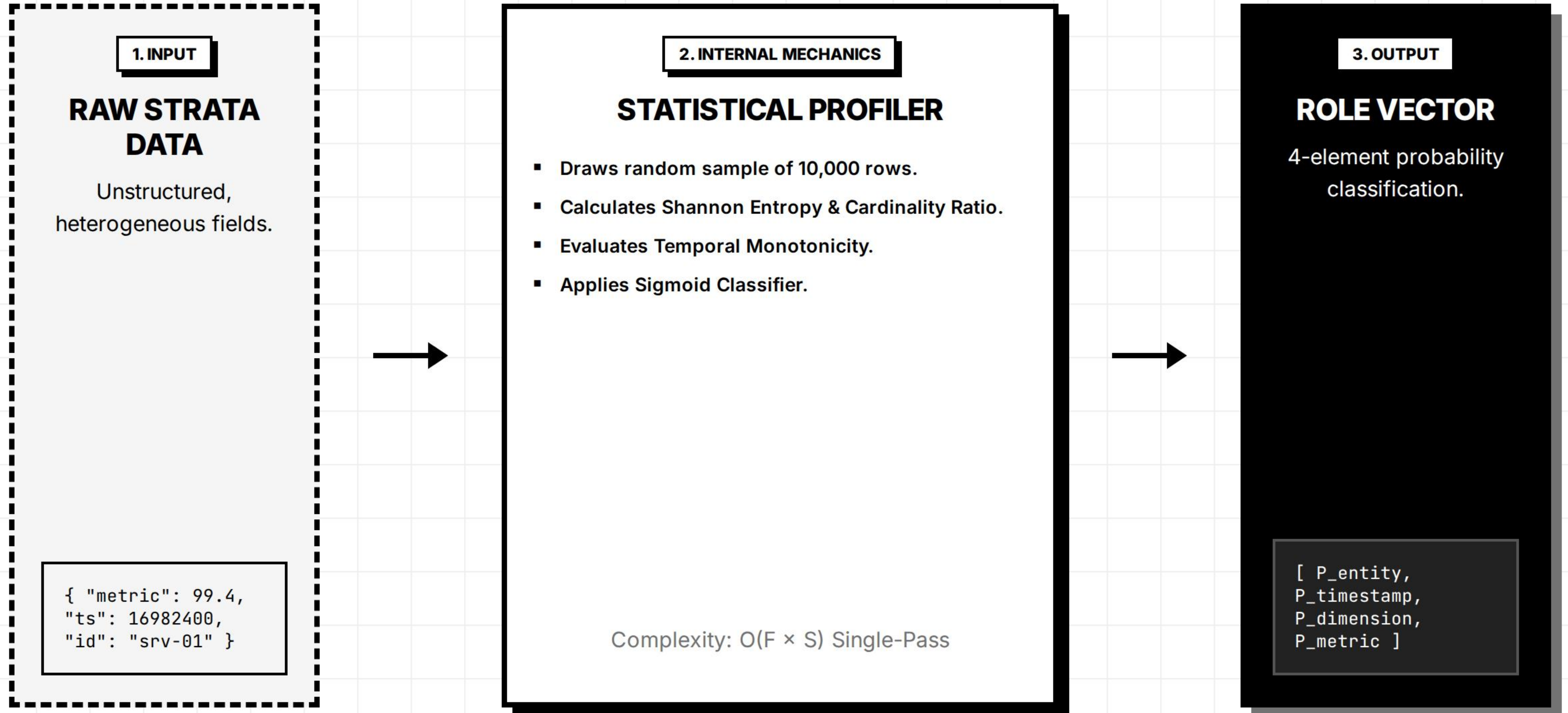
Autonomous Multi-Stratum Anomaly Correlation Engine.

Detailing Internal Mechanics, Mathematical Algorithms, and Data Passing Protocols for all 8 Modules (Version 2.0).

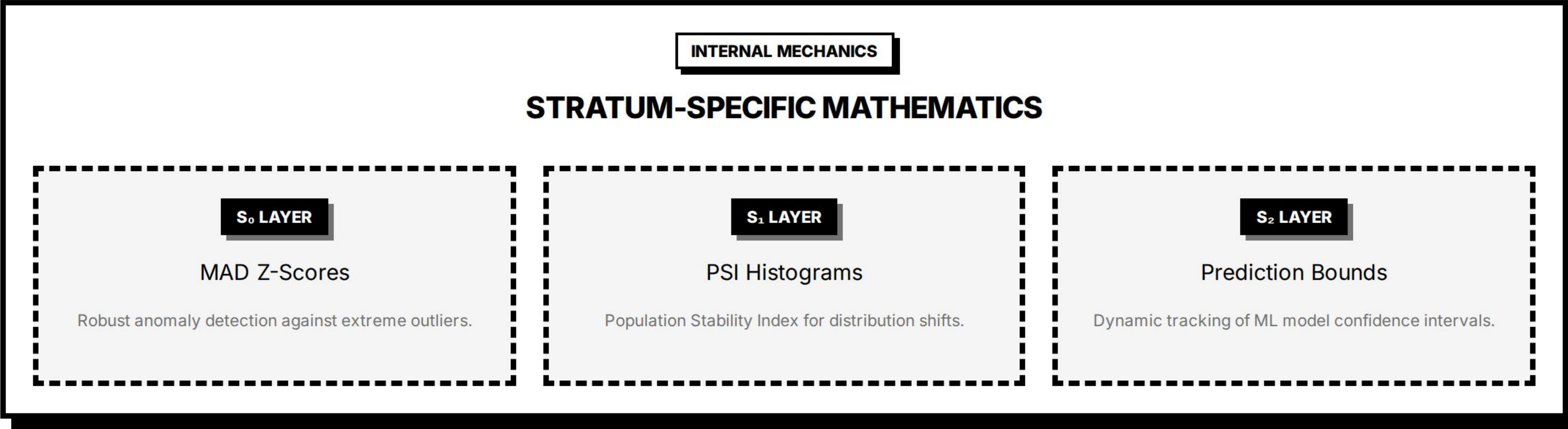
SYSTEM ARCHITECTURE & DATA FLOW



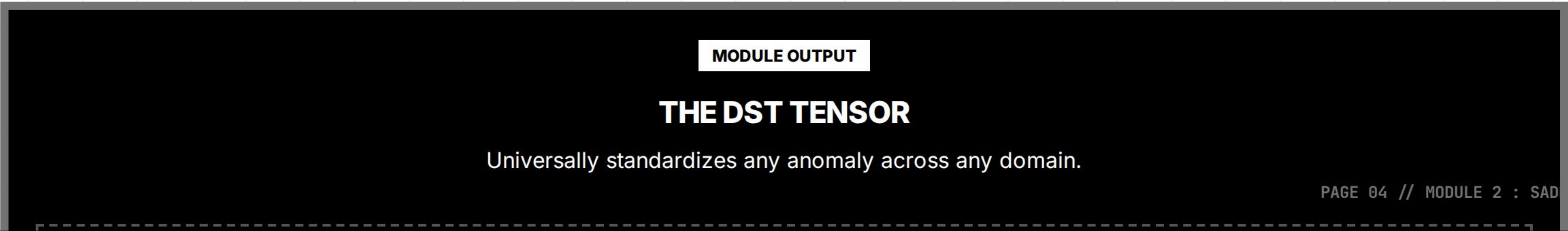
M1: SCHEMA INFERENCE ENGINE



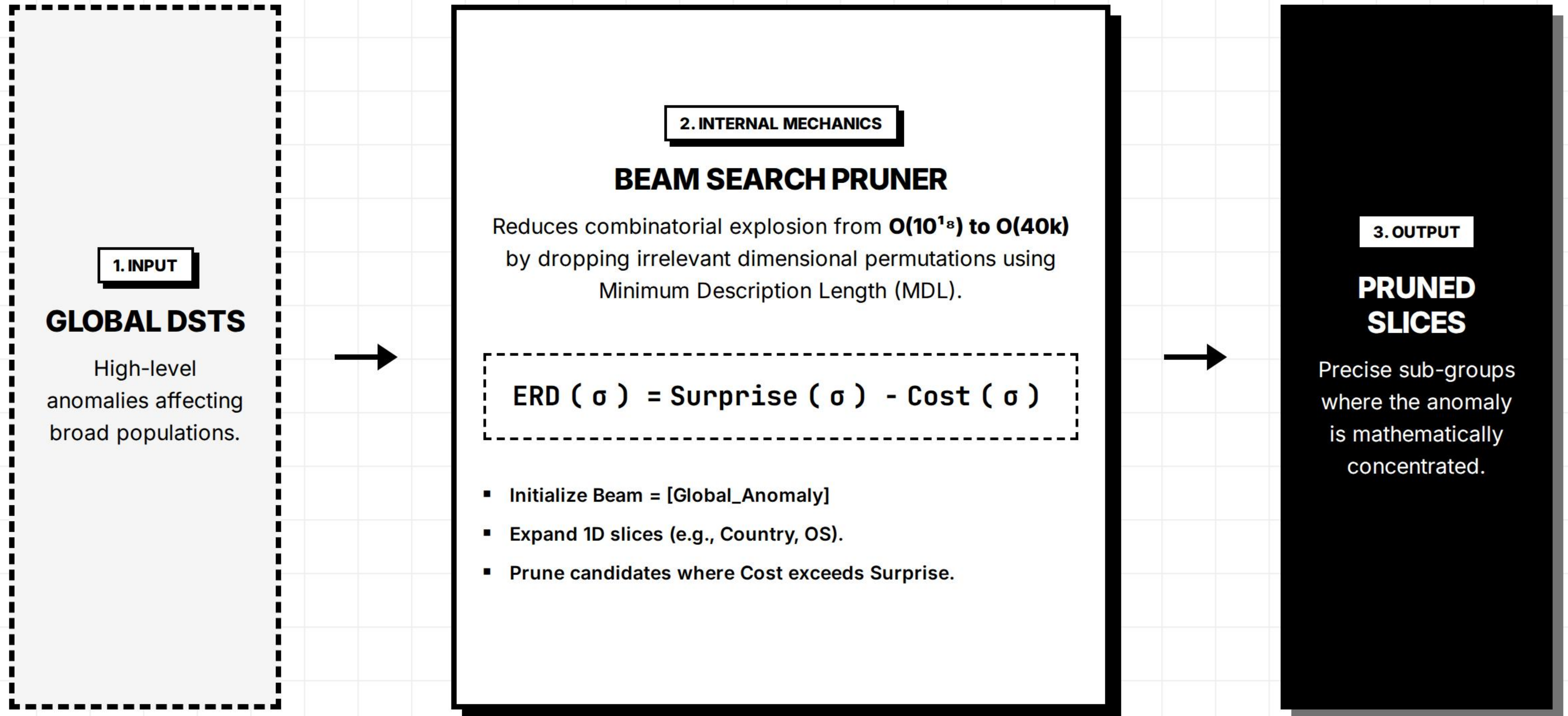
M2: ADAPTIVE DEVIATION DETECTOR



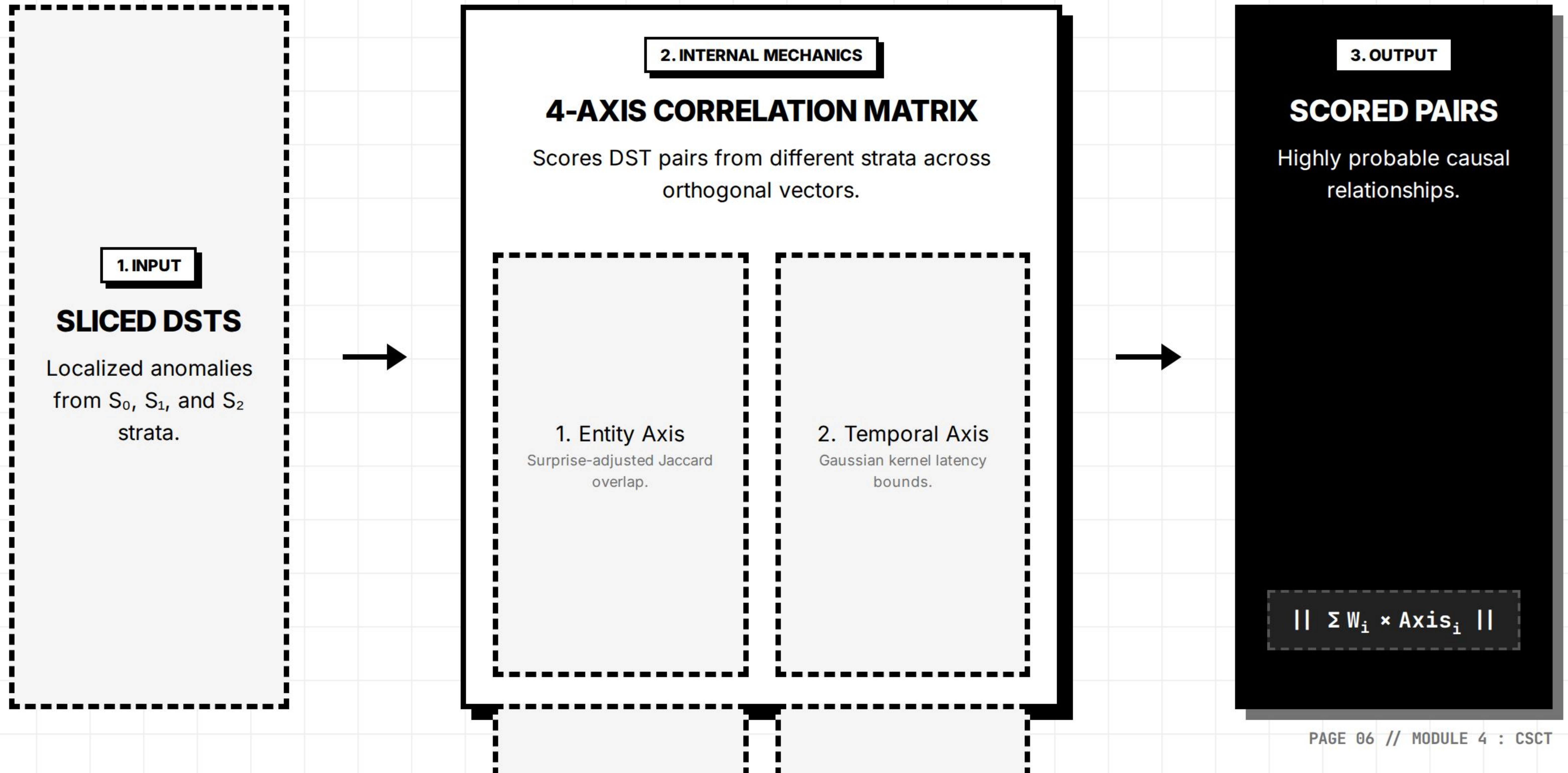
Unifies into Standard Format



M3: ENTROPIC RELEVANCE (ERD)



M4: CROSS-STRATUM CAUSAL TENSOR



M5: COMPUTATIONAL ABLATION

THE CAUSAL VALIDATION ENGINE

1000X BOOTSTRAP COUNTERFACTUALS

Executes physical data removal, recomputation, and bootstrapped CI tests at 95% confidence bounds.
Proves actual causality rather than coincidental correlation.

TEST 1

TEST 2

TEST 3

OUTPUT: Mathematically Validated Causal Links (Forwarded to M6, M7, M8)

M6-M8: SYNTHESIS & PREDICTION

MODULE 6

CIGS

Causal Insight Graphs

- DAG construction.
- Topological root finding.
- Entity set intersection.

Output: Root Cause Story

MODULE 7

PPE

Predictive Precursors

- Sequence matching.
- Stored historical patterns.
- Temporal decay math.

Output: Warning Alerts

MODULE 8

CIL

Contextual Insights

- Hybrid RAG Retrieval.
- Constrained LLM Prompt.
- 4-Layer Hallucination Check.

Output: Org Context